

Cornell University

Merton's Default Risk Framework: Empirical and Theoretical Application of Option Pricing Theory on Credit Risk Market

ORIE 5610: Stochastic Calculus II Project

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1 Background and Motivation

Background of Credit Risk Market

Finance rests on promises to repay: borrowers raise funds; lenders supply liquidity and earn compensation for time value, liquidity, and the risk that those promises are not met in full and on time. Credit risk therefore permeates large debt markets—for example, U.S. Treasuries (about \$28.3 T, ~60% of total) and U.S. corporate bonds (about \$11.2 T, ~24%)—and shapes how capital is allocated, balance sheets evolve, and macro-financial stability is tested when obligations mature, roll over, or default together.

Challenge for Quantifying Default Risk

Lenders and investors must judge whether a counterparty can and will service debt under stress. That judgment must be quantitative: outcomes are uncertain, path-dependent, and sensitive to macro conditions, collateral, covenants, funding, and other creditors. Mis-measured or concentrated risk can turn small shocks into large, nonlinear losses that propagate across institutions.

Motivation for using Stochastic Calculus

Stochastic processes model uncertain cash flows and default timing—e.g., shocks to earnings, rates, and asset prices as diffusions or jump-diffusions, and default as first passage or hazard-driven jumps. Stochastic calculus supports arbitrage-free pricing of credit-sensitive claims, hedging when payoffs depend on paths and factors, and dynamic risk measures rather than static spreadsheets. It does not eliminate default; it formalizes uncertainty, interaction, and the timing of losses—gaps laid bare when correlated shocks hit structured credit in 2008.

Goal

This project uses stochastic calculus as a framework for modeling and analyzing credit risk.

2 Merton's 1974 Credit Default Risk Framework

2.1 Economic setup: balance sheet and default

Let A_t denote the *market* value of the firm's assets (unlevered firm value). Claims are split between equity E_t and a *single* zero-coupon debt B_t that promises a fixed payment D at maturity T (face value / principal). The static accounting identity is

$$A_t = E_t + B_t, \quad (1)$$

where B_t is the *market value of debt* at time t . At maturity T , limited liability for equity means shareholders never pay in additional cash: they walk away if assets cannot cover debt.

Default is defined as

$$\text{default at } T \iff E_T \leq 0 \iff A_T < D. \quad (2)$$

2.2 Payoffs at maturity: European options on firm assets

Let $(x)^+ := \max(x, 0)$. At T , debt holders receive the smaller of promised principal or available assets; equity receives the residual:

$$\text{Debt (total payment to bondholders)} \quad B_T = \min(A_T, D) = D - (D - A_T)^+, \quad (3)$$

$$\text{Equity} \quad E_T = (A_T - D)^+. \quad (4)$$

Derivation: Equity $\equiv$ long European call on A_T

Define a European *call* on the underlying asset value with strike $K = D$ and expiry T :

$$\text{Call}_T = (A_T - D)^+.$$

Thus **shareholders hold a long European call** on firm assets with strike equal to outstanding debt face value:

$$E_T = \max(A_T - D, 0). \quad (5)$$

Intuition: if $A_T > D$, they “exercise” economic control and keep $A_T - D$ after paying debtholders D . If $A_T \leq D$, the call expires worthless and equity is zero—limited liability caps downside at zero while keeping upside convex in A_T .

Derivation: Debt $\equiv$ promised cash flow minus (short) put protection sold to equity

Rewrite the debtholder payoff:

$$B_T = D - \max(D - A_T, 0). \quad (6)$$

The term $\max(D - A_T, 0)$ is the maturity payoff of a European *put* on assets with strike D . Therefore

$$B_T = \underbrace{D}_{\text{promised principal}} - \underbrace{(D - A_T)^+}_{\text{payoff of a long put on } A \text{ with } K=D}. \quad (7)$$

Equivalently, relative to owning the full promised amount D for sure, debtholders **forfeit default losses** $(D - A_T)^+$ when $A_T < D$. That forfeiture is exactly what the *writer* of a European put pays (the short-put position at maturity). So debtholders are economically **long a riskless claim to D** and **short a European put** on firm assets with strike D (sometimes phrased as: risky debt = safe debt minus a put on assets). **Consistency check:** add claims:

$$E_T + B_T = (A_T - D)^+ + (D - (D - A_T)^+) = A_T,$$

using the identity $(x)^+ - (-x)^+ = x$.

Summary

We discovered that equity and debt valuation problem can be solved via european call and put option pricing .

3 European Option Pricing Models under Merton's Framework

We will discuss 4 different pricing models with the focus on call option pricing. Note that we can always recover the put option price via call-put parity, thus the debt value.

Put option (put-call parity).

$$p_t = c_t + K e^{-r(T-t)} - A_t. \quad (8)$$

Continuous Stochastic Dynamic for assets

Assume the asset value follows geometric Brownian motion under the physical measure (for modeling dynamics),

$$\frac{dA_t}{A_t} = \mu(A_t, t) dt + \sigma(A_t, t) dW_t, \quad (9)$$

with constant parameters in the simplest case $\mu(A_t, t) \equiv \mu$, $\sigma(A_t, t) \equiv \sigma$.

Valuation under no arbitrage (FTAP)

Under standard structural assumptions—complete markets, no dividends on A_t , constant volatility σ , constant short rate r , and that A_t is tradable (or a spanning argument holds)—the Fundamental Theorem of Asset Pricing implies existence of an equivalent martingale measure under which the *discounted* asset price is a martingale and claims are priced by expectation.

3.1 Black-Scholes-Merton

Then today's values are the discounted risk-neutral expectations of maturity payoffs. Using Black-Scholes notation $c(\cdot)$ and $p(\cdot)$ for European call and put prices on underlying spot S_t with strike K , maturity T , volatility σ , rate r ,

$$E_t = c^{\text{BS}}(S_t = A_t, K = D, T, \sigma, r), \quad (10)$$

$$B_t = D e^{-r(T-t)} - p^{\text{BS}}(S_t = A_t, K = D, T, \sigma, r), \quad (11)$$

where $D e^{-r(T-t)}$ is the time- t value of a zero-coupon bond paying D at T .

With the close form

$$c^{\text{BS}}(S_0, K, T, r, \sigma) = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2),$$

$$p^{\text{BS}}(S_0, K, T, r, \sigma) = K e^{-rT} \Phi(-d_2) - S_0 \Phi(-d_1),$$

with

$$d_1 = \frac{\ln(S_0/K) + \left(r + \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}},$$

$$d_2 = d_1 - \sigma\sqrt{T} = \frac{\ln(S_0/K) + \left(r - \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}}.$$

This model will be served as the benchmark model for comparison.

3.2 Heston model

Under an equivalent martingale measure $\mathbb{Q}$, let the spot S_t and its instantaneous variance $v_t \geq 0$ satisfy

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \\ dv_t &= \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^{(2)}, \\ d\langle W^{(1)}, W^{(2)} \rangle_t &= \rho dt \end{aligned} \quad (12)$$

with parameters $r \geq 0$ (short rate), $\kappa > 0$ (mean-reversion speed of variance), $\theta > 0$ (long-run variance level), $\xi > 0$ (vol-of-vol), and correlation $\rho \in [-1, 1]$. Initial conditions are $S_0 > 0$ and $v_0 \geq 0$.

European call: Fourier form

Consider a European call with strike K and maturity T . Heston's semi-closed form expresses the call as

$$c_0^{\text{Heston}} = S_0 P_1 - K e^{-rT} P_2, \quad (13)$$

where each P_j is a *tail probability* computed from a characteristic function ϕ_j :

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left(\frac{e^{-iu \ln K} \phi_j(u)}{iu} \right) du, \quad j = 1, 2. \quad (14)$$

The put option price can be obtained by call-put parity

Characteristic functions

Define

$$\phi_j(u) = \exp(C_j(u, T) + D_j(u, T)v_0 + iu \ln S_0), \quad (15)$$

where complex-valued auxiliary functions C_j, D_j depend on (u, T) and the model parameters. Set

$$d_j = \sqrt{(\rho \xi i u - b_j)^2 - \xi^2 (2\alpha_j i u - u^2)}, \quad (16)$$

$$g_j = \frac{b_j - \rho \xi i u + d_j}{b_j - \rho \xi i u - d_j}, \quad (17)$$

using the principal branch for the square root in implementations. Then

$$C_j(u, T) = iurT + \frac{\kappa\theta}{\xi^2} \left[(b_j - \rho \xi i u + d_j)T - 2 \ln \left(\frac{1 - g_j e^{d_j T}}{1 - g_j} \right) \right], \quad (18)$$

$$D_j(u, T) = \frac{b_j - \rho \xi i u + d_j}{\xi^2} \left(\frac{1 - e^{d_j T}}{1 - g_j e^{d_j T}} \right). \quad (19)$$

The index-dependent constants are

$$\alpha_1 = \frac{1}{2}, \quad \alpha_2 = -\frac{1}{2}, \quad b_1 = \kappa - \rho \xi, \quad b_2 = \kappa. \quad (20)$$



Figure 1: Stock price chart of Credit Suisse in 2023

3.3 Jump Diffusion model

Motivation for discontinuous dynamics for assets

Continuous diffusion models cannot reproduce the sudden vertical price moves, called "Jump". A concrete illustration is the behavior of *Credit Suisse* (NYSE: CS) around the banking stress of March 2023: after a relatively stable stretch in late 2022–early 2023, the price fell abruptly from on the order of a few dollars to below one dollar, with a pronounced spike in trading volume at the same time.

To capture the jump, we need to introduce a count process to randomly introduce a jump event and a size distribution whenever a jump happened.

3.3.1 Poisson process

A (homogeneous) *Poisson process* $(N_t)_{t \geq 0}$ with *rate* $\lambda > 0$ is a càdlàg counting process with $N_0 = 0$, independent increments, and for all $0 \leq s < t$,

$$N_t - N_s \sim \text{Poisson}(\lambda(t - s)).$$

Equivalently, one may think of jumps of size 1 at random event times, with interarrival times independent and identically distributed as $\text{Exp}(\lambda)$.

Moments. For each fixed $t \geq 0$,

$$\mathbb{E}[N_t] = \lambda t, \quad \text{Var}(N_t) = \lambda t.$$

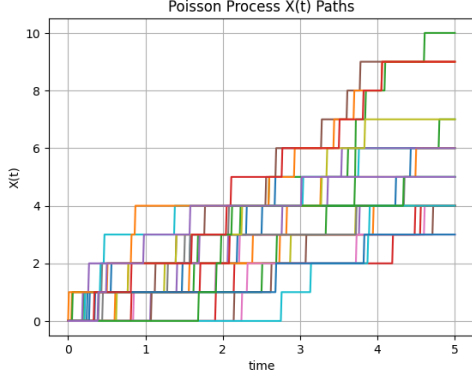


Figure 2: Sample paths of a standard Poisson process

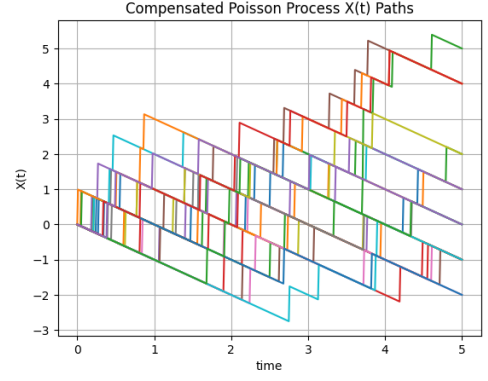


Figure 3: Sample paths of a compensated Poisson process

Compensated Poisson process

Let $(N_t)_{t \geq 0}$ be a homogeneous Poisson process with rate $\lambda > 0$, and let $(\mathcal{F}_t)_{t \geq 0}$ denote its natural filtration.

Define the *compensated Poisson process*

$$M_t := N_t - \lambda t. \quad (21)$$

The process $(M_t)_{t \geq 0}$ is a martingale with respect to $(\mathcal{F}_t)$. Heuristically, λt is the predictable “drift” that centers the counting process so that the upward jumps of N_t are offset between jump times; this is the discrete-jump analogue of the centered (Brownian) part and is central in stochastic integration with respect to jump processes.

Compound Poisson process

Let $(Y_n)_{n \geq 1}$ be i.i.d. real random variables, independent of (N_t) . The *compound Poisson process* is

$$X_t := \sum_{n=1}^{N_t} Y_n, \quad X_t := 0 \text{ when } N_t = 0. \quad (22)$$

Between jumps of N , X_t is constant; at each jump time of N , X increases (or decreases, if Y_n is negative) by an independent increment Y_n .

Compensated compound Poisson process

Assume in addition that $\mathbb{E}[|Y_n|] < \infty$. Then, conditioning on N_t ,

$$\mathbb{E}[X_t] = \mathbb{E}[N_t] \mathbb{E}[Y_1] = \lambda t \mathbb{E}[Y_1]. \quad (23)$$

The *compensated compound Poisson process* is defined by

$$\tilde{X}_t := X_t - \lambda t \mathbb{E}[Y_1] = \sum_{n=1}^{N_t} Y_n - \lambda t \mathbb{E}[Y_1]. \quad (24)$$

Then $(\tilde{X}_t)_{t \geq 0}$ is a martingale with respect to the natural filtration generated by (N_t) and the jump sizes (Y_n) .

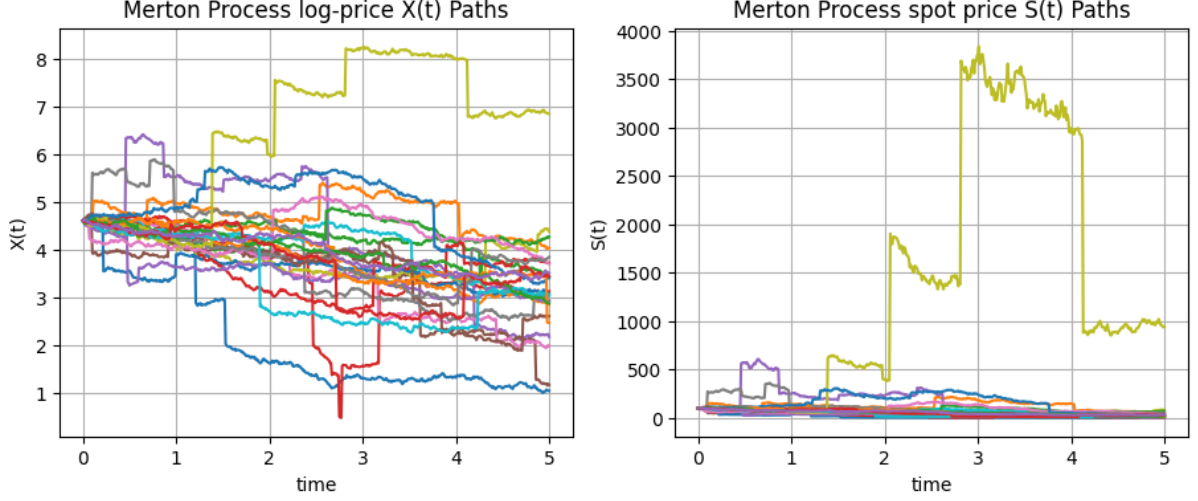


Figure 4: Merton's Jump Diffusion Process

3.3.2 Merton's Jump diffusion dynamic

Assume the asset price A_t follows Merton's jump-diffusion process:

$$\frac{dA_t}{A_t} = (\mu - \lambda k) dt + \sigma dW(t) + d\left(\prod_{i=1}^{N_t} J_i - 1\right), \quad (25)$$

where:

- $W(t)$ is a standard Brownian motion,
- $N_t \sim \text{Poisson}(\lambda t)$ is the jump-counting process,
- $J_i \sim LN(\mu_J, \sigma_J^2)$ are i.i.d. jump multipliers with $k = E[J_i - 1]$,
- λk is the compensator drift correction, so A_t is a local martingale under the risk-neutral measure when $\mu = r$.

It can be rewritten as

$$\frac{dA_t}{A_t} = (\mu - \lambda k)dt + \sigma dW(t) + \sum_{n \in \mathbb{N}} (J_n - 1) \mathbf{1}(\Gamma_n = t) \quad (26)$$

where:

- Z_i is defined to be exponentially distributed, denote the time needed for the next jump,
- $\Gamma(n) = \sum_{i=1}^n Z_i$ denotes the arrival time of n-th arrival,

3.3.3 Itô's Lemma for Jump-Diffusion Processes

Let $f(A) = \log A$. For a jump-diffusion process the generalised Itô formula is

$$df(A_t) = f'(A_t) dA_t^c + \frac{1}{2} f''(A_t) (dA_t^c)^2 + [f(A_{t-} J_{N_t}) - f(A_{t-})] dN_t, \quad (27)$$

where $dA_t^c = (\mu - \lambda k)A_t dt + \sigma A_t dW(t)$ is the continuous part.

Computing each term with $f'(A) = 1/A$ and $f''(A) = -1/A^2$:

$$f'(A_t) dA_t^c = \frac{1}{A_t} [(\mu - \lambda k) A_t dt + \sigma A_t dW(t)] = (\mu - \lambda k) dt + \sigma dW(t), \quad (28)$$

$$\frac{1}{2} f''(A_t) (dA_t^c)^2 = \frac{1}{2} \left(-\frac{1}{A_t^2} \right) \sigma^2 A_t^2 dt = -\frac{1}{2} \sigma^2 dt, \quad (29)$$

$$f(A_{t-} J_{N_t}) - f(A_{t-}) = \log(A_{t-} J_{N_t}) - \log(A_{t-}) = \log J_{N_t}. \quad (30)$$

Substituting back:

$$d \log A_t = \left(\mu - \lambda k - \frac{1}{2} \sigma^2 \right) dt + \sigma dW(t) + \log J_{N_t} dN_t. \quad (31)$$

3.3.4 Asset Density under Jump Model

Integrating from 0 to T and summing the jump contributions:

$$\log \frac{A_T}{A_0} = \left(\mu - \lambda k - \frac{1}{2} \sigma^2 \right) T + \sigma W(T) + \sum_{i=1}^{N_T} \log J_i, \quad (32)$$

so that

$$A_T = A_0 \exp \left[\left(\mu - \lambda k - \frac{1}{2} \sigma^2 \right) T + \sigma W(T) \right] \prod_{i=1}^{N_T} J_i. \quad (33)$$

Under the risk-neutral measure, set $\mu = r$ throughout (and keep λk as the compensator):

$$A_T = A_0 \exp \left[\left(r - \lambda k - \frac{1}{2} \sigma^2 \right) T + \sigma W(T) \right] \prod_{i=1}^{N_T} J_i. \quad (34)$$

Lognormal Jumps (Merton's Specification)

Following Merton (1976), assume $\log J_i \sim \mathcal{N}(\gamma, \delta^2)$ so that

$$k = E[J_i - 1] = e^{\gamma + \frac{1}{2} \delta^2} - 1, \quad (35)$$

$$E[\log(1 + \varepsilon_i)] = \gamma - \frac{1}{2} \delta^2, \quad \text{Var}[\log(1 + \varepsilon_i)] = \delta^2, \quad (36)$$

where $\varepsilon_i = J_i - 1$.

Conditioning on $N_T = n$

Conditional on $N_T = n$, the sum $\sum_{i=1}^n \log J_i \sim \mathcal{N}(n\gamma, n\delta^2)$. Therefore, conditional on $N_T = n$, the log-return is normally distributed:

$$\log \frac{A_T}{A_0} \Big| N_T = n \sim \mathcal{N} \left(\left(r - \lambda k - \frac{1}{2} \sigma^2 \right) T + n\gamma, \sigma^2 T + n\delta^2 \right). \quad (37)$$

This means $A_T \mid N_T = n$ is lognormally distributed, exactly as required by the Black–Scholes model, but with adjusted parameters:

$$\sigma_n^2 = \sigma^2 + \frac{n\delta^2}{T}, \quad r_n = r - \lambda k + \frac{n\gamma}{T}. \quad (38)$$

To see this, note the conditional log-variance over $[0, T]$ must equal $\sigma_n^2 T$:

$$\sigma_n^2 T = \sigma^2 T + n\delta^2 \implies \sigma_n^2 = \sigma^2 + \frac{n\delta^2}{T}. \quad (39)$$

Similarly, the conditional mean of $\log A_T/A_0$ must equal $(r_n - \frac{1}{2}\sigma_n^2)T$:

$$\left(r - \lambda k - \frac{1}{2}\sigma^2\right)T + n\gamma = \left(r_n - \frac{1}{2}\sigma_n^2\right)T \quad (40)$$

$$\implies r_n = r - \lambda k + \frac{n\gamma}{T}. \quad (41)$$

Since $\gamma = \log(1+k)$ in Merton's parameterisation, this is consistent with the notation $r_n = r + \frac{n\gamma}{T} - \lambda k$ shown in the text.

Conditional distribution given $N_T = n$. From

$$\log\left(\frac{A_T}{A_0}\right) = \left(r - \lambda k - \frac{1}{2}\sigma^2\right)T + \sigma W_T + \sum_{i=1}^{N_T} \log J_i,$$

and Merton's assumption $\log J_i \sim \mathcal{N}(\gamma, \delta^2)$, we have

$$\sum_{i=1}^n \log J_i \sim \mathcal{N}(n\gamma, n\delta^2).$$

Hence, conditional on $N_T = n$,

$$\log\left(\frac{A_T}{A_0}\right) \mid N_T = n \sim \mathcal{N}\left(\left(r - \lambda k - \frac{1}{2}\sigma^2\right)T + n\gamma, \sigma^2T + n\delta^2\right).$$

Define adjusted Black-Scholes parameters

$$\sigma_n^2 := \sigma^2 + \frac{n\delta^2}{T}, \quad r_n := r - \lambda k + \frac{n\gamma}{T}.$$

Then equivalently

$$\log\left(\frac{A_T}{A_0}\right) \mid N_T = n \sim \mathcal{N}\left(\left(r_n - \frac{1}{2}\sigma_n^2\right)T, \sigma_n^2T\right),$$

so

$$A_T \mid N_T = n \sim \text{Lognormal}\left(\log A_0 + \left(r_n - \frac{1}{2}\sigma_n^2\right)T, \sigma_n^2T\right).$$

Therefore, conditional on $N_T = n$, A_T has the same form as in Black-Scholes with parameters (r_n, σ_n) .

Marginal (unconditional) distribution

Unconditionally,

$$A_t = \sum_{n=0}^{\infty} \underbrace{\frac{(\lambda T)^n}{n!} e^{-\lambda T}}_{\Pr(N_T=n)} \cdot \mathbb{P}[A_t \mid N_T = n] \quad (42)$$

3.3.5 Merton's European Call Option Price

The risk-neutral price of a European call with strike D and maturity T is

$$C^{\text{JD}} = e^{-rT} E^{\mathbb{Q}}[\max(A_T - D, 0)]. \quad (43)$$

Apply the tower property, conditioning on N_T :

$$C^{\text{JD}} = e^{-rT} \sum_{n=0}^{\infty} P(N_T = n) E^{\mathbb{Q}}[\max(A_T - D, 0) \mid N_T = n]. \quad (44)$$

Because $A_T \mid N_T = n$ is lognormal with parameters (σ_n^2, r_n) , the inner expectation is exactly the undiscounted Black–Scholes call payoff with those parameters. Absorbing the e^{-rT} discount into each BS term:

$$E^{\mathbb{Q}}[\max(A_T - D, 0) \mid N_T = n] \cdot e^{-rT} = e^{-(r_n - r_n + r)T} (\dots) = C^{\text{BS}}(A_t, D, T, \sigma_n^2, r_n), \quad (45)$$

where C^{BS} is the standard Black–Scholes call price:

$$C^{\text{BS}}(A_t, D, T, \sigma_n^2, r_n) = A_t \Phi(d_1) - D e^{-r_n T} \Phi(d_2), \quad (46)$$

$$d_1 = \frac{\log(A_t/D) + (r_n + \frac{1}{2}\sigma_n^2)T}{\sigma_n \sqrt{T}}, \quad d_2 = d_1 - \sigma_n \sqrt{T}. \quad (47)$$

The Poisson PMF is

$$P(N_T = n) = \frac{(\lambda T)^n}{n!} e^{-\lambda T}. \quad (48)$$

However, because the compensated jump arrival rate under the risk-neutral measure shifts $\lambda \rightarrow \lambda' = \lambda(1 + k)$ (to preserve the martingale property of the discounted price process), we replace $\lambda \rightarrow \lambda'$:

$$P(N_T = n) = \frac{(\lambda' T)^n}{n!} e^{-\lambda' T}, \quad \lambda' = \lambda(1 + k). \quad (49)$$

Putting everything together:

European Call Price as an Infinite Sum of Black–Scholes Prices

$$C^{\text{JD}}(A_t, D, T, \sigma^2, r, \delta^2, \lambda, k) = \sum_{n=0}^{\infty} \frac{(\lambda' T)^n}{n!} e^{-\lambda' T} C^{\text{BS}}(A_t, D, T, \sigma_n^2, r_n), \quad (50)$$

with

$$\lambda' = \lambda(1 + k), \quad (51)$$

$$r_n = r - \lambda k + \frac{n\gamma}{T}, \quad (52)$$

$$\sigma_n^2 = \sigma^2 + \frac{n\delta^2}{T}, \quad (53)$$

$$\gamma = \log(1 + k). \quad (54)$$

Note: Although formula (50) involves an infinite series, but numerical evaluation is fast: typically only the first 10–15 terms are needed for most practical applications.

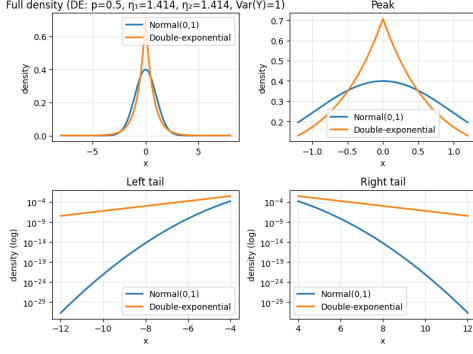


Figure 5: symmetric Laplace density

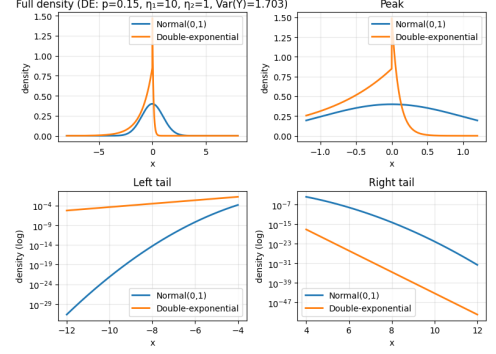


Figure 6: asymmetric Laplace density

3.4 Kou's Jump Diffusion model

One main issue about Merton's model is the choice of log-normally distributed jump size, it failed to capture the leverage effect (i.e. market respond more aggressively to negative market shock) and leptokurtic distribution (i.e. heavy tail, to capture extreme event). Kou (2002) proposed asymmetric Laplacian distribution as alternative.

3.4.1 Gaussian Distribution vs Double Exponential Distribution

The key innovation in Kou's model is that $Y = \log(J)$ has an **asymmetric double exponential distribution** (also called the Laplacian distribution) with probability density function:

$$f_X(x) = p\eta_1 e^{-\eta_1 x} \mathbf{1}_{\{x \geq 0\}} + q\eta_2 e^{\eta_2 x} \mathbf{1}_{\{x < 0\}}, \quad 0 < \eta_1, \eta_2 < \infty, \quad (55)$$

where $p, q \geq 0, p + q = 1$, and $\eta_1 > 1, \eta_2 > 0$.

Alternative representation: The double exponential distribution can be written as

$$Y = \begin{cases} \xi & \text{with probability } p, \\ -\xi & \text{with probability } q, \end{cases} \quad (56)$$

where $\xi \sim \text{Expo}(1/\eta)$ with mean η and variance η^2 , with density

$$f(\xi) = \frac{1}{\eta} e^{-\xi/\eta}, \quad 0 < \xi < \infty. \quad (57)$$

Equivalently, we can write:

$$\log(J_i) = Y = \begin{cases} \xi_+ & \text{with probability } p, \\ -\xi_- & \text{with probability } q, \end{cases} \quad (58)$$

where ξ_+ and ξ_- are exponential random variables with means $1/\eta_1$ and $1/\eta_2$, respectively.

As Figure 5 and 6 shown, with the appropriate setting of the laplace distribution, we can obtain a left heavy tails and right thin massed density, producing the leverage effect and Lep-tokurtic feature.

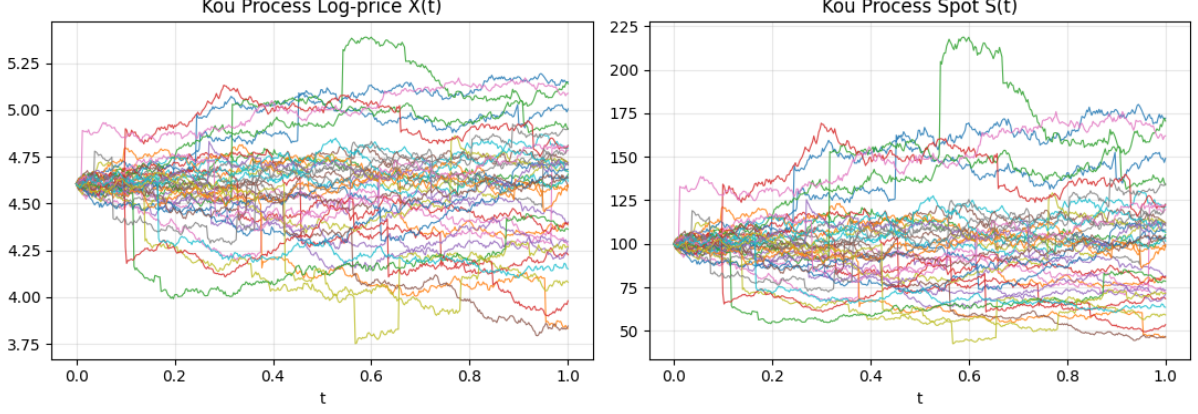


Figure 7: Kou's Jump diffusion Process

Properties of the Double Exponential Distribution

Some useful properties:

$$E[Y] = \kappa = \frac{p}{\eta_1} - \frac{q}{\eta_2} = \frac{p}{\eta_1} - \frac{1-p}{\eta_2}, \quad (59)$$

$$\text{Var}[Y] = 2\eta^2, \quad (60)$$

$$E[e^Y] = E[J] = \frac{e^\kappa}{1-\eta^2} = \frac{p\eta_1}{\eta_1-1} + \frac{q\eta_2}{\eta_2+1}. \quad (61)$$

The requirement $\eta_1 > 1$ ensures that $E[J] < \infty$ and $E[A_t] < \infty$, meaning the average upward jump cannot exceed 100%.

3.4.2 Kou's Jump Diffusion Dynamic

Assume the asset price A_t follows Kou's jump-diffusion process under the physical measure:

$$\frac{dA_t}{A_{t-}} = \mu dt + \sigma dW_t + d\left(\sum_{i=1}^{N_t} (J_i - 1)\right), \quad (62)$$

where:

- W_t is a standard Wiener process (Brownian motion),
- $N_t \sim \text{Poisson}(\lambda t)$ is the jump-counting process with rate λ ,
- $J_i \sim \text{Laplace}(p, \eta_1, \eta_2)$ are i.i.d. jump multipliers, with $k = E[J_i - 1]$,
- n_t , W_t , and $\{J_i\}$ are independent.

3.4.3 Asset Density under Kou's model

Integrating the dynamics gives the closed-form level (Kou's equation for S_t):

$$A_t = A_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma W_T\right] \prod_{i=1}^{N_T} J_i = A_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma W_T + \sum_{i=1}^{N_T} Y_i\right], \quad (63)$$

Note that $\prod_{i=1}^0 J_i = 1$ by convention (when there are no jumps).

Conditional distribution given $N_T = n$. Fix $n \in \{0, 1, 2, \dots\}$. Conditional on $N_T = n$, the Wiener increment W_T and the n jump logs $\{Y_i\}_{i=1}^n$ are independent, hence

$$\log \frac{A_t}{A_0} \Big| (N_T = n) = \left(\mu - \frac{1}{2} \sigma^2 \right) T + \sigma W_T + \sum_{i=1}^n Y_i. \quad (64)$$

Equivalently, writing $Z \sim \mathcal{N}(0, 1)$,

$$\log \frac{A_t}{A_0} \Big| (N_T = n) \stackrel{d}{=} \left(\mu - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z + \sum_{i=1}^n Y_i. \quad (65)$$

Relation to the Merton (lognormal-jump) case.

- If $n = 0$, then $\sum_{i=1}^0 Y_i = 0$ and

$$\log \frac{A_t}{A_0} \Big| (N_T = 0) \sim \mathcal{N}\left(\left(\mu - \frac{1}{2} \sigma^2\right)T, \sigma^2 T\right),$$

so $A_t \mid (N_T = 0)$ is *exactly lognormal* (pure diffusion).

- If $n \geq 1$, the jump contribution $\sum_{i=1}^n Y_i$ is *not* Gaussian under the double-exponential law for Y , so in general $\log(A_t/A_0) \mid (N_T = n)$ is *not* normal and therefore $A_t \mid (N_T = n)$ is *not* lognormal (unlike Merton's Gaussian log-jumps).

3.4.4 Characteristic function of Kou's model

Instead of dealing with the convolution of the poisson-log-normal mixture distribution, Kou has derived the characteristic function:

Let $\psi_Y(u) = \mathbb{E}[e^{iuY}]$ be the characteristic function of one jump log-size. Then, conditional on $N_T = n$,

$$\mathbb{E}\left[\exp\left(iu \log \frac{A_t}{A_0}\right) \Big| N_T = n\right] = \exp\left(iu\left(\mu - \frac{1}{2} \sigma^2\right)T - \frac{1}{2} \sigma^2 T u^2\right) (\psi_Y(u))^n, \quad (66)$$

which characterizes the conditional distribution of $\log(A_t/A_0)$ for all n .

3.4.5 Kou's Jump process Option pricing model

3.5 Risk-Neutral Dynamics

In the presence of random jumps, the market becomes incomplete and standard hedging arguments are not applicable. However, assuming jump risk is diversifiable, we can derive an option pricing formula under a risk-neutral measure $\mathbf{Q}$ such that

$$\begin{aligned} \frac{dA_t}{A_t} &= [r - \lambda E(J - 1)] dt + \sigma dW_t + d\left(\sum_{i=1}^{n_t} (J_i - 1)\right) \\ &= (r - \lambda \psi) dt + \sigma dW_t + d\left(\sum_{i=1}^{n_t} (J_i - 1)\right), \end{aligned} \quad (67)$$

where r is the risk-free interest rate, $J = e^X$ such that X follows the double exponential distribution with density f_X given in Eq.,

$$\psi = E[J - 1] = E[e^X] - 1 = \frac{e^\kappa}{1 - \eta^2} - 1, \quad 0 < \eta < 1, \quad (68)$$

and the parameters $\kappa, \eta, \psi, \sigma$ become risk-neutral parameters taking consideration of risk premiums.

Risk-Neutral Asset Price

The unique solution of the risk-neutral dynamics is

$$A_t = A_0 \exp \left[\left(r - \frac{\sigma^2}{2} - \lambda\psi \right) t + \sigma W_t \right] \prod_{i=1}^{n_t} J_i. \quad (69)$$

European Call Option Price

The price of a European call option at time t is given by

$$c_t = E_* \left[e^{-r(T-t)} (A_t - K)_+ \right], \quad (70)$$

where $(y)_+ = \max(0, y)$, K is the strike price, T is the expiration time, and the expectation is taken under the risk-neutral measure $\mathbf{Q}$. Explicitly:

$$c_t = E^{\mathbf{Q}} \left[e^{-r(T-t)} \left(A_t \exp \left[\left(r - \frac{\sigma^2}{2} - \lambda\psi \right) (T-t) + \sigma \sqrt{T-t} \epsilon \right] \prod_{i=1}^{n_T} J_i - K \right)_+ \right], \quad (71)$$

where ϵ is a standard normal random variable.

Analytical Pricing Formula

Kou (2002) shows that c_t is analytically tractable as

$$c_t = \sum_{n=1}^{\infty} \sum_{j=1}^n e^{-\lambda(T-t)} \frac{\lambda^n (T-t)^n}{n!} \frac{2^j}{2^{2n-1}} \binom{2n-j-1}{n-1} (A_{1,n,j} + A_{2,n,j} + A_{3,n,j}) \\ + e^{-\lambda(T-t)} \left[A_t e^{-\lambda\psi(T-t)} \Phi(h_+) - K e^{-r(T-t)} \Phi(h_-) \right], \quad (72)$$

where $\Phi(\cdot)$ is the CDF of the standard normal random variable and

$$A_{1,n,j} = A_t e^{-\lambda\psi(T-t) + n\kappa} \frac{1}{2} \left(\frac{1}{(1-\eta)^j} + \frac{1}{(1+\eta)^j} \right) \Phi(b_+) - e^{-r(T-t)} K \Phi(b_-), \quad (73)$$

$$A_{2,n,j} = \frac{1}{2} e^{-r(T-t) - \omega/\eta + \sigma^2(T-t)/(2\eta^2)} K \sum_{i=0}^{j-1} \left(\frac{1}{(1-\eta)^{j-i}} - 1 \right) \left(\frac{\sigma\sqrt{T-t}}{\eta} \right)^i \frac{1}{\sqrt{2\pi}} Hh_i(c_-), \quad (74)$$

$$A_{3,n,j} = \frac{1}{2} e^{-r(T-t) + \omega/\eta + \sigma^2(T-t)/(2\eta^2)} K \sum_{i=0}^{j-1} \left(1 - \frac{1}{(1+\eta)^{j-i}} \right) \left(\frac{\sigma\sqrt{T-t}}{\eta} \right)^i \frac{1}{\sqrt{2\pi}} Hh_i(c_+), \quad (75)$$

with auxiliary quantities

$$b_{\pm} = \frac{\ln(A_t/K) + (r \pm \sigma^2/2 - \lambda\psi)(T-t) + n\kappa}{\sigma\sqrt{T-t}}, \quad (76)$$

$$h_{\pm} = \frac{\ln(A_t/K) + (r \pm \sigma^2/2 - \lambda\psi)(T-t)}{\sigma\sqrt{T-t}}, \quad (77)$$

$$c_{\pm} = \frac{\sigma\sqrt{T-t}}{\eta} \pm \frac{\omega}{\sigma\sqrt{T-t}}, \quad (78)$$

$$\omega = \ln(K/A_t) + \lambda\psi(T-t) - (r - \sigma^2/2)(T-t) - n\kappa, \quad (79)$$

$$\psi = \frac{e^{\kappa}}{1 - \eta^2} - 1. \quad (80)$$

The Hh Functions

The $Hh_n(\cdot)$ functions are defined by

$$Hh_n(x) = \frac{1}{n!} \int_x^\infty (s-x)^n e^{-s^2/2} ds, \quad n = 0, 1, 2, \dots, \quad (81)$$

with boundary cases $Hh_{-1}(x) = e^{-x^2/2}$ (which equals $\sqrt{2\pi} \phi(x)$, where ϕ is the standard normal PDF) and $Hh_0(x) = \sqrt{2\pi} \Phi(-x)$.

The Hh_n functions satisfy the three-term recursion

$$n Hh_n(x) = Hh_{n-2}(x) - x Hh_{n-1}(x), \quad n \geq 1, \quad (82)$$

so all values for $n \geq 1$ can be computed from the normal density ϕ and normal CDF Φ alone.

Limiting Cases

- If $\lambda = 0$ (no jumps), formula (72) reduces to the standard Black–Scholes call price.
- As $\eta_1 \rightarrow \infty$ and $\eta_2 \rightarrow \infty$ (jump sizes shrink to zero), the formula again converges to Black–Scholes.

Note: Although formula (72) involves an infinite series, numerical evaluation is fast: typically only the first 10–15 terms are needed for most practical applications.

4 Application to Finance

This chapter we discuss the application of the models we just introduced to finance in terms of 1) Producing volatility smile, 2) Delta Hedging, 3) Option pricing

4.1 Producing volatility smile

We know that Black-Scholes-Merton model assume a flat volatility surface, thus we focus on the jump model.

4.1.1 Merton's Jump Diffusion model

Recall the jump part of Merton's jump diffusion process:

$$\prod_{i=1}^{N_t} J_i - 1$$

where:

- $N_t \sim \text{Poisson}(\lambda t)$ is the jump-counting process,
- $J_i \sim LN(\mu_J, \sigma_J^2)$ are i.i.d. jump multipliers

There are 3 hyperparameter that have different effect onto the volatility surface. By setting default parameter $\sigma_J = 0.2$, $\lambda_p = 0.1$, $\mu_J = 0$, we get:

The takeaway is when there is more volatile jumps, the volatility surface will have a "deeper" smile. We also noticed that when σ_J and λ_p equal to 0, the volatility surface becomes flat (i.e. no jump, BS setting).

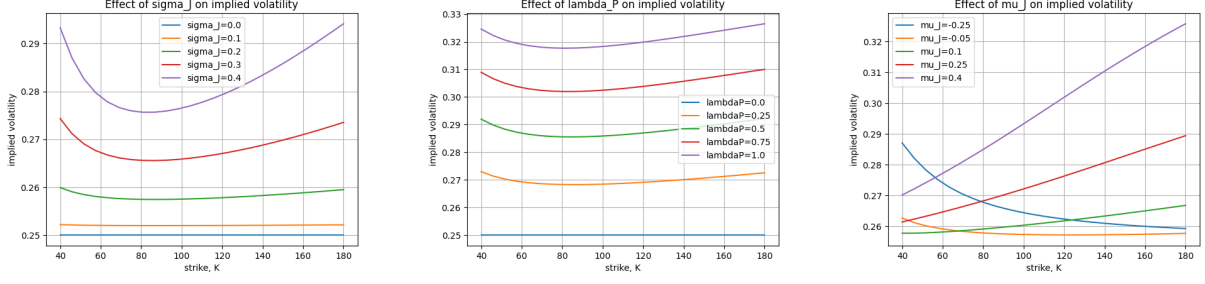


Figure 8: Effect of Merton's jump parameter (volatility, intensity, mean) on IV

4.1.2 Kou's Jump Diffusion model

Recall the jump part of Merton's jump diffusion process:

$$\prod_{i=1}^{N_t} J_i - 1$$

where:

- $N_t \sim \text{Poisson}(\lambda t)$ is the jump-counting process,
- $\log(J_i) \sim \text{Laplace}(\eta_1, \eta_2, p)$ are i.i.d. jump multipliers

There are 4 hyperparameter that have different effect onto the volatility surface. By setting default parameter $\lambda = 0.5$, $\eta_1 = 10$, $\eta_2 = 10$, $p = 0.5$, we get:

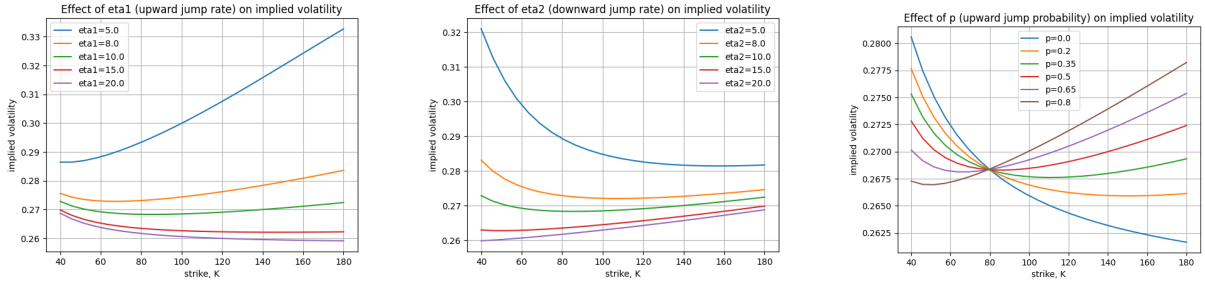


Figure 9: Effect of Kou's jump parameter (η_1 , η_2 , p) on IV

The takeaway is that volatility smile can be adjusted to be a smirk with appropriate setting (e.g. lower η_2 can produce a left smirk according to plot). We also noticed that when η_1 and η_2 approach to ∞ , the volatility surface becomes flat (i.e. no jump, BS setting).

4.1.3 Comparison

There are 2 issues from Merton's model:

- Hard to distinguish the source of volatility smile
- Hard to produce the volatility smirk

Figure 8 showed that σ_J and λ_P produce very similar shapes and both failed to produce the volatility smirk that is often observed in the real market. Hence, Kou's model can theoretically fit the market better than Merton's model.

4.2 Delta Hedging performance for jump model

We apply the usual Black-Scholes-Merton delta hedging scheme to the Jump process and evaluate their running performance:

- Number Of Paths = 1000
- Number Of Steps = 2000
- $T = 1.0$
- $r = 0.1$
- $\sigma = 0.2$
- $\lambda_P = 3.0$
- $\mu_J = 0.0$
- $\sigma_J = 0.25$
- $p_{kou} = 0.5$
- $\eta_1 = 10$
- $\eta_2 = 5$
- $s_0 = 1.0$
- $K = 0.95$ (Vanilla call option)

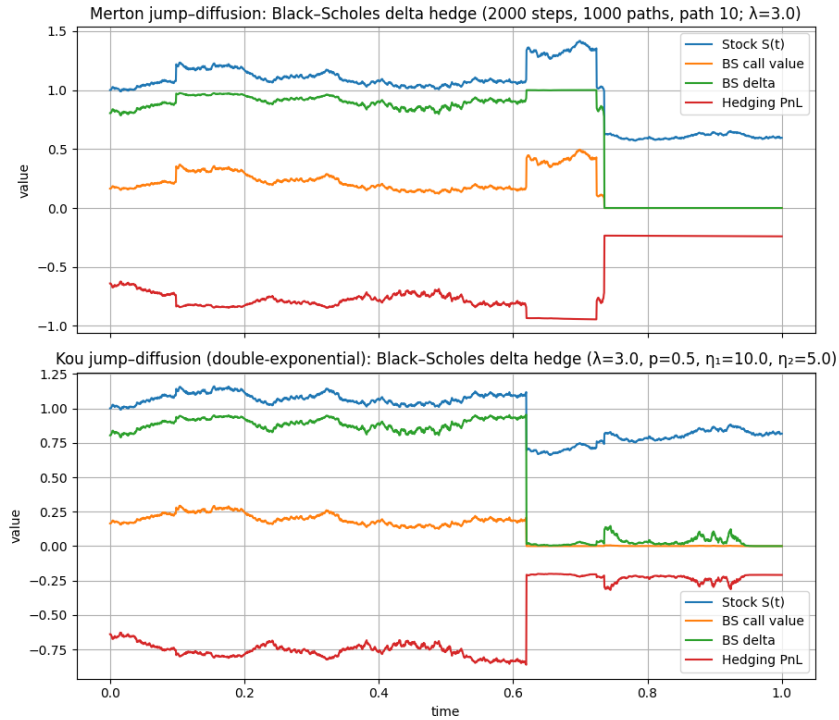


Figure 10: Running PnL and delta hedging performance plot of single path

We noticed that the delta hedging can't capture the sudden jump of the process, thus resulted into a negative PnL.

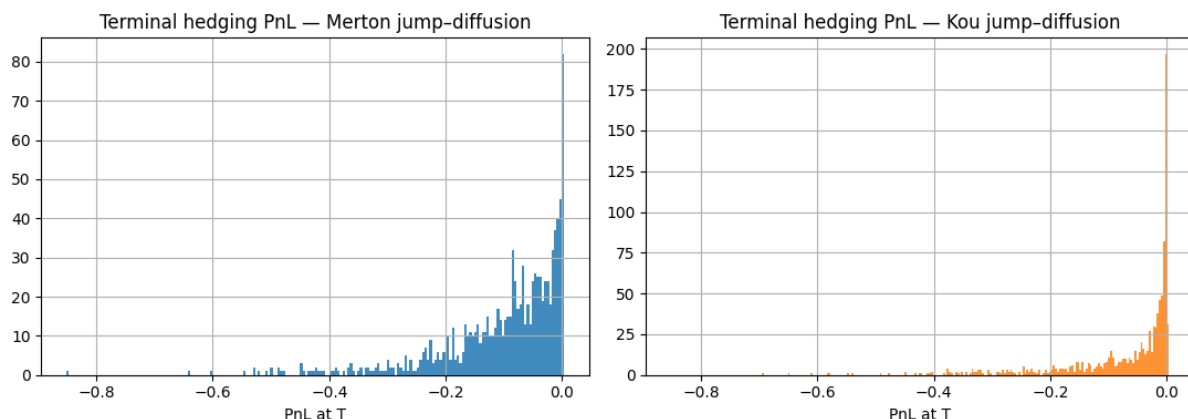


Figure 11: Histogram of delta hedging PnL: 1000 simulations

As Figure 10 suggested, the PnL are all negative. Another takeaway is that the PnL have a higher peak at 0^- in Kou's model compared to Merton's model, suggesting that the Kou's model can yield a better delta hedging performance in most cases in terms of risk natural pricing.

4.2.1 Hedging Jump

According to Kou (2002), he argued that the hedging for the jump part is negligible, since perfect delta hedging is almost impossible in discrete time, thus there is always imperfect hedging in practice even if there is no jump. Tsay (2010) and Oosterlee (2019) claimed that jump risk can be hedged by diversification, it involved constructing a portfolio of assets with the same jump source in opposite position with appropriate choice of weight. Further studies required.

4.2.2 Jump Effect on Pricing

There are 2 issues with pricing model with jump:

- Market incompleteness: Even though the risk-neutral measure exist ,but since the jump is not perfectly hedgeable, therefore there will be **no unique risk-neutral pricing**.
- Memoryless: Both jump models doesn't have memory of the previous jump due to the memoryless properties of poisson process. To reproduce the conditional heteroskedasticity, one may consider a dependent jump count process or fractional brownian motion.

4.3 Theoretical pricing comparison

Table 1: European **call** prices: OTM vs. ITM ($K = 81$; OTM when $S_0 < K$, ITM when $S_0 > K$).

Model	OTM call ($S_0 = 80$)	ITM call ($S_0 = 82$)
Black–Scholes	3.49	4.69
Heston (QuantLib)	3.47	4.64
Merton jump-diffusion	3.78	4.98
Kou jump-diffusion	3.92	5.11

Common: $K = 81$, $T = 0.25$, $r = 0.08$, $\sigma = 0.2$.

Heston: $v_0 = 0.04$, $\kappa = 0.1$, $\theta = 0.04$, $\sigma_v = 0.1$, $\rho = 0.7$.

Merton: $\Lambda = 10$, $\mu_J = -0.02$, $\sigma_J = 0.02$, $N = 50$ series terms.

Kou: $\lambda = 10$, $\kappa = \mathbb{E}(Y) = \frac{p}{\eta_1} - \frac{1-p}{\eta_2} = -0.02$, $\eta_1 = \eta_2 = 0.02$.

Table 2: European **put** prices: OTM vs. ITM ($K = 81$; OTM when $S_0 > K$, ITM when $S_0 < K$).

Model	OTM put ($S_0 = 82$)	ITM put ($S_0 = 80$)
Black–Scholes	2.08	2.89
Heston (QuantLib)	2.04	2.87
Merton jump-diffusion	2.38	3.18
Kou jump-diffusion	2.51	3.31

Common: $K = 81$, $T = 0.25$, $r = 0.08$, $\sigma = 0.2$.

Heston / Merton / Kou: same as Table 1.

Higher prices under Merton/Kou than BS at the same σ mean the market assigns substantial weight to large discrete moves; differences between Merton and Kou mean tail shape and compensator matter as much as jump intensity λ — different jump laws are not interchangeable even when λ is shared.

4.4 Implementation Note

In the Merton structural setup the *value of the firm* (total assets) A_t and its dynamics are not directly traded or continuously observed: public investors typically see the equity price and, at best, (quarterly) balance-sheet information, while the true asset process remains a latent state. In practice one therefore *approximates* the current asset level by combining market observables for equity with book or market values for debt using the *fundamental accounting identity* (assets equal claims on the firm),

$$E_t = (\text{shares outstanding}) \times S_t, \quad (83)$$

$$A_t = E_t + D_t. \quad (84)$$

where E_t is the market value of equity (e.g. shares outstanding times the stock price S_t) and D_t is the face value or market value of debt as specified in the implementation. Even with such a proxy for A_t , the *volatility of assets* σ_A is not quoted in markets; empirical work usually substitutes a measurable equity volatility proxy, such as *realised volatility* estimated from stock returns (as in this project) or *implied volatility* from traded options. Because these choices identify neither

A_t nor σ_A uniquely, the resulting prices are best interpreted as *relative valuations* across names or scenarios: cross-sectional comparisons are on firmer ground when all entities are calibrated with the *same* volatility and balance-sheet mapping convention, so that differences in outputs reflect model-implied credit/equity spreads rather than inconsistent data construction.

Additionally, the debt structure is also not publicly observable. For simplicity, we will assume the total debt value in the balance sheet 1) is a constant, 2) share the same maturity date/payment structure, and 3) there is no coupons. For the multiple debt payment structure and coupons setup, refer to appendix (7.1).

In summary, I will impose the following assumption for the empirical study:

- Time t : Dec 31, 2025
- Asset process: $A_t = \text{Market Cap} - D = \# \text{ Outstanding share} * S_t - D$
- Asset Volatility: $\sigma_A \approx \sigma_{IV}^{ATM}$
- Liability value: $D = \text{Reported total liability (Constant)}$

5 Empirical result

We will focus on 3 main metrics to empirically evaluate the model:

- Credit valuation (Equity & Debt)
- Credit Spread
- Probability of Default (PD)

5.1 Stock Selection and Corresponding Credit Risk Profile

We select six publicly-traded firms spanning the credit quality spectrum to examine how model choice affects pricing outcomes under varying default risk regimes. The sample is deliberately constructed to contrast investment-grade names with distressed entities:

(Safe) Investment-Grade Firms. We study three liquid, low-leverage firms: Apple (AAPL), Tesla (TSLA), and Oracle (ORCL). They have high asset-to-debt ratios ($A_0/K \gg 1$; AAPL ≈ 40), moderate ATM implied volatilities (30–60%), and a deeply out-of-the-money Merton default put. Hence default risk is minimal, so all four models should yield similar debt prices.

(Risky) Distressed Firms. We also study three high-leverage, high-volatility firms: GameStop (GME), AMC (AMC), and Bed Bath & Beyond (BBBY, which filed for Chapter 11 bankruptcy in 2023). Their asset-to-debt ratios are near 1 (BBBY ≈ 1.1), ATM implied volatilities are very high (often 80–100%; GME 94%), and the Merton default put is near or in the money. Thus default risk is material, and model choice—especially downside-jump dynamics—should meaningfully affect debt prices and implied credit spreads.

This split lets us test whether richer dynamics improve on BSM when default risk is high, while confirming convergence across models when default risk is remote.

5.2 Hyperparameter Calibration

For simplicity, we pre-determined default value for the hyperparameter instead of market data calibration.

- **BSM (baseline):**
 - Asset dynamics: geometric Brownian motion with risk-free drift r .
 - r from 3-month U.S. Treasury yield.
 - σ is ticker-specific ATM IV.
- **Heston stochastic volatility:**
 - Parameters: $\kappa = 2.0$, $\theta = \sigma^2$, $v_0 = \sigma^2$, $\xi = 0.30$, $\rho = -0.70$.
 - Numerics: Monte Carlo with 20,000 paths, 252 steps/year, Euler full-truncation (Andersen, 2008).
- **Merton jump-diffusion:**
 - Jump parameters: $\Lambda = 1.0$, $\mu_J = -0.05$, $\sigma_J = 0.10$.
 - Pricing: equity via truncated Merton series (50 terms); debt via call-put parity.
- **Kou double-exponential jump-diffusion:**
 - Jump parameters: $\lambda = 2.0$, $p = 0.4$, $\eta_1 = 10.0$, $\eta_2 = 5.0$.
 - Interpretation: more frequent jumps, asymmetric tails (60% down-jump probability), small up-jumps vs larger down-jumps.
 - Pricing: equity via Gil-Pelaez Fourier inversion ($u_{\max} = 200$); debt via put-call parity.

5.3 Credit Valuation

The valuation here refers to relative valuation, we will study the value of equity and debt under the Merton's default risk framework. Note that the fair valuation is not possible to obtain under the constraint of data quality and data source (refer to section 4.4 for details)

5.3.1 Equity Pricing

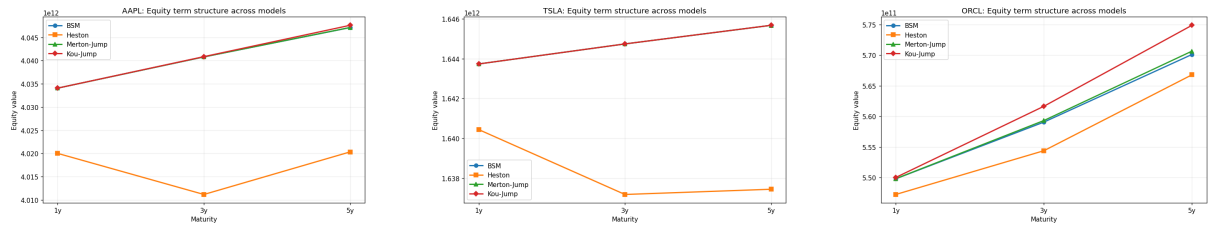


Figure 12: Various equity pricing scheme for investment-grade firms

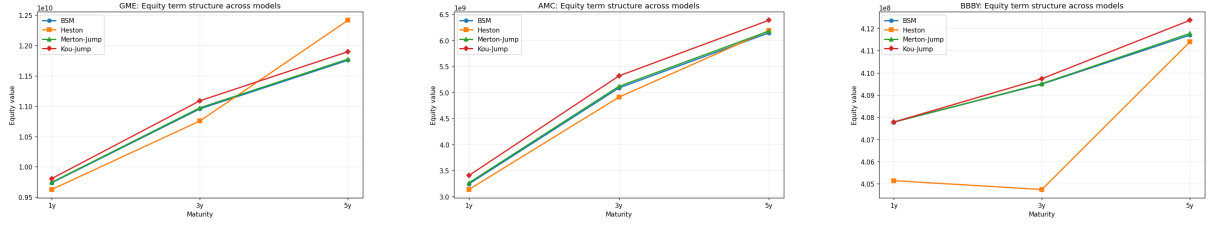


Figure 13: Various equity pricing scheme for distressed firms

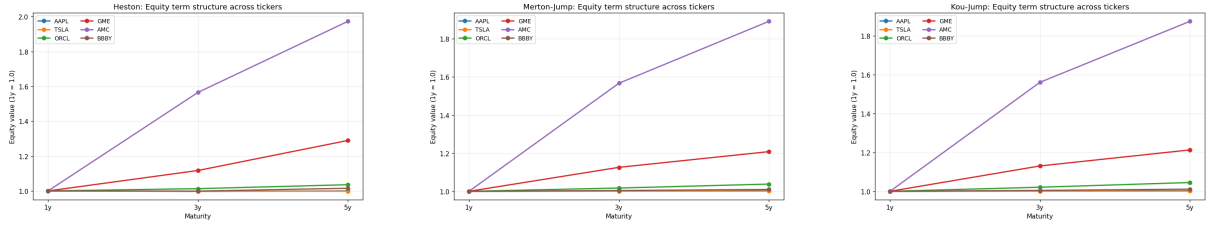


Figure 14: Various equity pricing scheme for different firms (Normalized to 1y value)

We observed that equity pricing under jump model are higher than BSM model, matching our theoretical derivation in section 4.3. Kou's jump model yields a higher price than Merton's jump model due to the heavy left tail. The pricing under Heston model is more unpredictable as it doesn't exhibit the same shape as the other models, possible cause is the insufficient steps (20000) for the monte carlo simulation when used Euler's scheme.

5.3.2 Debt Pricing

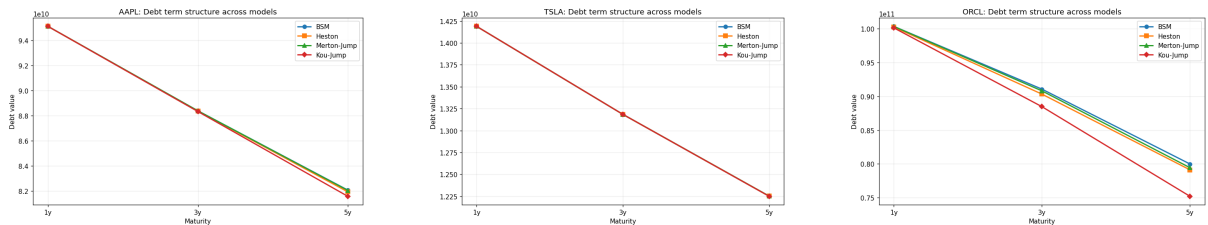


Figure 15: Various debt pricing scheme for investment-grade firms

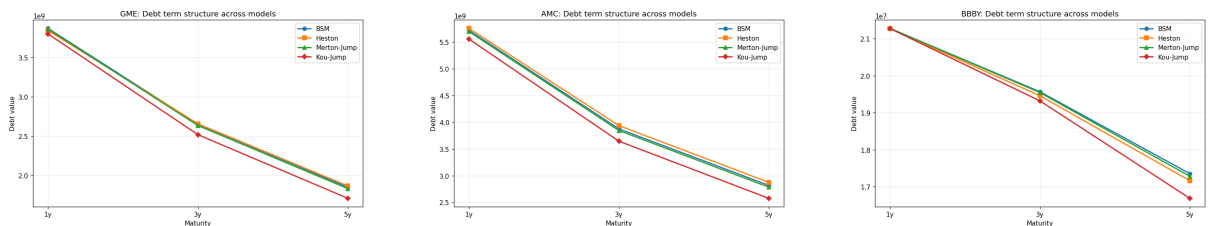


Figure 16: Various debt pricing scheme for distressed firms

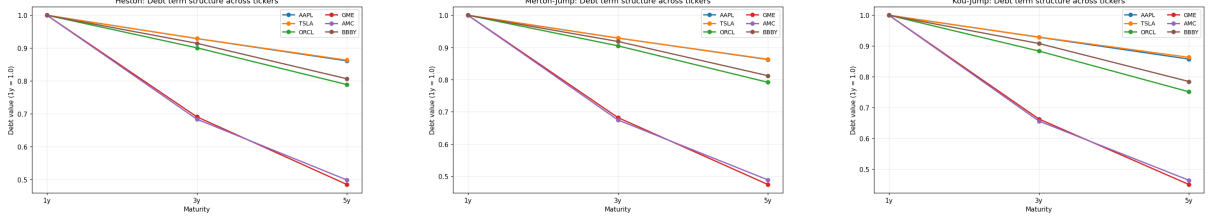


Figure 17: Various pricing scheme for different firms (Normalized to 1y value)

The pricing in debt plots are similar to the pricing in equity plot but in upside down, which intuitively make sense since debt value can be viewed as a short position of a put option. The put options are deep OTM for investment-grade firms, therefore, their debt value is very stable whereas distressed assets debt value drop more as T grow, matched out assumption at the beginning.

5.3.3 Takeaway for Model Selection

Our results suggest practitioners should employ jump-diffusion models (particularly Kou's asymmetric specification) when pricing debt for speculative-grade credits with leverage ratios $A_0/K < 2$ and maturities exceeding 3 years. For investment-grade firms ($A_0/K > 5$) or short-dated debt ($T < 1$ year), the simpler BSM framework suffices. Heston stochastic volatility offers only marginal improvements over BSM unless one is pricing exotic credit derivatives (e.g., quanto CDS, volatility-linked notes) where smile dynamics matter explicitly.

5.4 Credit Spread

Definition and Economic Interpretation The *credit spread* $s(T)$ is the additional yield an investor demands for bearing default risk relative to a risk-free bond of identical maturity. Formally, if a risky bond with face value F trades at price B_T (defined = corporate bond = debt), its yield to maturity is

$$y_{\text{risky}}(T) = -\frac{1}{T} \ln \frac{B_T}{D},$$

while the risk-free yield is r . The credit spread is their difference:

$$s(T) = y_{\text{risky}}(T) - r = -\frac{1}{T} \ln \frac{B_T}{De^{-rT}}.$$

Economically, $s(T)$ compensates investors for the expected loss given default (LGD) and the probability of default (PD) over the horizon $[0, T]$. Under the risk-neutral measure, the spread also incorporates a risk premium for bearing jump and volatility risk.

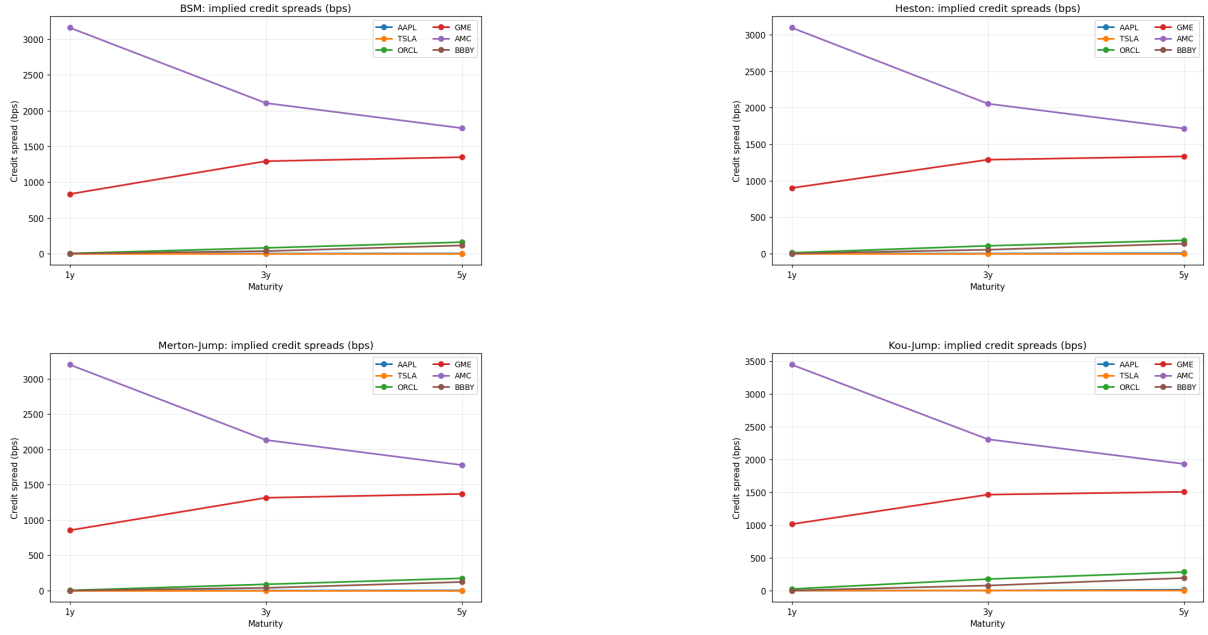


Figure 18: Implied credit spread under different models

We noticed the investment-grade firms are almost flat at 0 spread, matching with their credit rating that investing in them is close to be default-risk free. Whereas the spread is wide for distressed firms, Kou's heavy left tail indicated the widest spread amongst all models. The spread exhibits a tendency of convergence.

Note that the risky corporate BBY is also in the zone of safety asset (almost zero credit spread), possible reason for that are 1) the chapter 11 bankruptcy require the company to reconstruct the debt structure such that it is "almost impossible" to default anymore **on paper**, 2) the model simply can't capture it's riskiness.

5.5 Probability of default (PD)

The one-period (horizon- T) default probability is the probability that default occurs by time T . Under the physical measure,

$$PD = \mathbb{P}(A_T < D) = \mathbb{P}(\text{default by } T),$$

and under the risk-neutral measure (for pricing),

$$PD^{\mathbb{Q}} = \mathbb{P}^{\mathbb{Q}}(A_T < D).$$

1. Black-Scholes-Merton (BSM)

Default probability.

$$PD(m) = \Phi(-d_2) = \Phi\left(-\frac{\log(A_0/D) + (m - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\right),$$

where $m = \mu$ (physical) or $m = r$ (risk-neutral).

2. Heston stochastic volatility

Asset dynamics.

$$\frac{dA_t}{A_t} = \mu dt + \sqrt{v_t} dW_1, \quad dv_t = \kappa(\theta - v_t) dt + \sigma_v \sqrt{v_t} dW_2,$$

with $dW_1 dW_2 = \rho dt$.

Characteristic function.

$$\phi_X(u) = \exp(C(u, T) + D^*(u, T) v_0),$$

where $X_T = \log(A_T/A_0)$ and C, D^* are the standard Heston characteristic-function coefficients.

Default probability (Gil–Pelaez).

$$\text{PD} = \mathbb{P}(A_T < D) = \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \frac{\text{Im} \left[e^{-iu \log(D/A_0)} \phi_X(u) \right]}{u} du.$$

3. Merton jump-diffusion

Log-asset return (conditional on $N_T = n$ jumps).

$$\log(A_T/A_0) | (N_T = n) \sim \mathcal{N} \left((\mu_n - \frac{1}{2}\sigma_n^2)T, \sigma_n^2 T \right),$$

where

$$\begin{aligned} \sigma_n^2 &= \sigma^2 + \frac{n\sigma_J^2}{T}, & \mu_n &= m - \lambda k + \frac{n\gamma}{T}, \\ k &= e^{\mu_J + \frac{1}{2}\sigma_J^2} - 1, & \gamma &= \log(1 + k). \end{aligned}$$

Default probability (Poisson mixture).

$$\text{PD} = \sum_{n=0}^{\infty} \mathbb{P}(N_T = n) \Phi(-d_{2,n}),$$

with

$$\mathbb{P}(N_T = n) = \frac{(\lambda' T)^n}{n!} e^{-\lambda' T}, \quad \lambda' = \begin{cases} \lambda & \text{(physical),} \\ \lambda(1 + k) & \text{(risk-neutral),} \end{cases}$$

and

$$d_{2,n} = \frac{\log(A_0/D) + (\mu_n - \frac{1}{2}\sigma_n^2)T}{\sigma_n \sqrt{T}}.$$

4. Kou jump-diffusion

Log-asset return characteristic function (Laplace jumps).

$$\phi_X(u) = \exp \left(iu\mu_x T - \frac{1}{2}\sigma^2 u^2 T + \lambda T (\phi_Y(u) - 1) \right),$$

where

$$\begin{aligned} \mu_x &= \mu - \frac{1}{2}\sigma^2 - \lambda\psi, & \psi &= \frac{e^\kappa}{1 - \eta^2} - 1, \\ \phi_Y(u) &= \frac{e^{iu\kappa}}{1 + \eta^2 u^2} \quad \text{(Laplace characteristic function).} \end{aligned}$$

Default probability. Apply Gil–Pelaez inversion,

$$\text{PD} = \mathbb{P}(A_T < D) = \mathbb{P}\left(X_T < \log \frac{D}{A_0}\right) = \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \frac{\text{Im}\left[e^{-iu \log(D/A_0)} \phi_X(u)\right]}{u} du,$$

$$\phi_X(u) = \exp\left(iu\mu_x T - \frac{1}{2}\sigma^2 u^2 T + \lambda T(\phi_Y(u) - 1)\right),$$

$$\mu_x = \mu - \frac{1}{2}\sigma^2 - \lambda\psi, \quad \psi = \frac{e^\kappa}{1 - \eta^2} - 1, \quad \phi_Y(u) = \frac{e^{iu\kappa}}{1 + \eta^2 u^2}.$$

5.5.1 Empirical Result

PD physical — all models ($T = 1$ year, $\mathbb{P}$)

Ticker	BSM	Heston	Merton	Kou
AAPL	1.025×10^{-31}	1.249×10^{-9}	1.154×10^{-13}	1.192×10^{-9}
BBBY	4.581×10^{-3}	8.620×10^{-3}	6.399×10^{-3}	4.642×10^{-3}
GME	2.032×10^{-1}	2.036×10^{-1}	2.364×10^{-1}	2.046×10^{-1}
ORCL	1.010×10^{-3}	3.854×10^{-3}	2.954×10^{-3}	1.057×10^{-3}

Table 3: One-year default probabilities by model: physical

The physical probability of default are all close to zero due to imperfect approximation of Merton’s framework. We shall focus on the comparison. The result is obvious that AAPL is relative unlikely to default and the GME is relatively likely to default. However, the model suggested that both GME and ORCL are more likely to default than a partially defaulted company BBBY, it might indicate a poor representation of reality for some high risk companies.

PD risk-neutral — all models ($T = 1$ year, $\mathbb{Q}$)

Ticker	BSM	Heston	Merton	Kou
AAPL	3.003×10^{-31}	1.253×10^{-9}	9.426×10^{-14}	1.182×10^{-9}
BBBY	6.683×10^{-3}	1.159×10^{-2}	8.741×10^{-3}	6.765×10^{-3}
GME	4.361×10^{-2}	5.639×10^{-2}	6.195×10^{-2}	4.439×10^{-2}
ORCL	1.929×10^{-3}	5.917×10^{-3}	4.606×10^{-3}	2.008×10^{-3}

Table 4: One-year default probabilities by model: risk-neutral measures

Here the risk neutral probability are all close to zero as well. Note that the risk neutral probability doesn’t carry any probabilistical meaning to our physical world, we shall focus on the physical measure in this context.

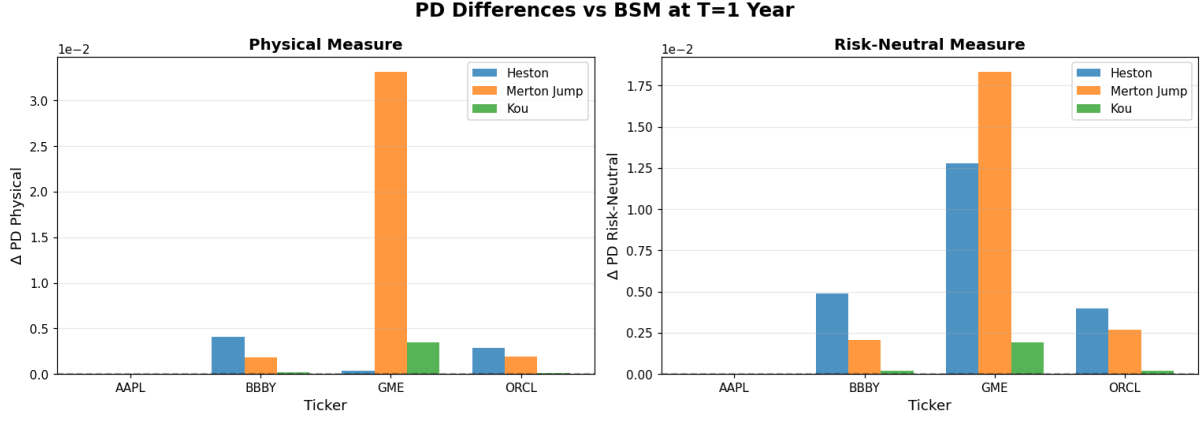


Figure 19: 1 years PD difference against the baseline BSM

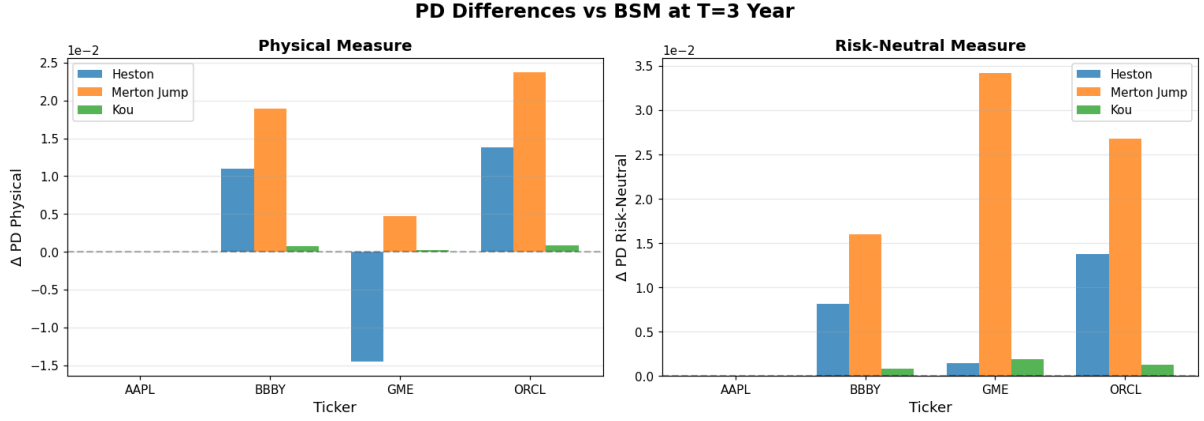


Figure 20: 3 years PD difference against the baseline BSM

The unit on the vertical axis shows that the difference between models are very small in terms of probability of default, suggesting that a simple model like BSM might be well suitable enough for such metric. Focus on the physical measure, the model suggest that GME is relatively likely to default in 1 year under Merton’s jump model, whereas ORCL and BBBY are more likely to default in 3 years under both Heston and Merton’s jump model. Note that Kou’s jump model is very consistent with the baseline BSM model (difference ≈ 0).

6 Conclusion

This report applied several stochastic-pricing engines within Merton’s structural setup to corporate equity and debt, linked those outputs to implied credit spreads, and computed the probability of default. The exercise complements the broader discussion of credit markets: model choice matters most when default risk is economically relevant, and much less when the credit option embedded in debt is effectively negligible.

The empirical results from the valuation notebook are mostly consistent with standard structural-credit intuition. For firms far from default—large asset-to-debt ratios $A_0/K \gg 1$ and moderate equity-implied volatility—debt behaves close to a risk-free claim with the same face value. In this mild-risk regime, the **Black–Scholes–Merton** (BSM) diffusion is typically sufficient for practical purposes across key outputs (credit valuation, implied credit spread, and PD). Because the embedded short default put is deep out of the money, refinements to tail

dynamics have limited pricing impact, so added model complexity often yields only marginal gains.

By contrast, for highly levered names or periods of elevated uncertainty—when A_0/K approaches unity and the equity option is nearer at the money—the default option becomes materially valuable, and model-implied left-tail behavior matters. In those settings, richer dynamics such as Heston stochastic volatility or jump-diffusion models can generate meaningfully different spreads and PDs from BSM. In our implementation, the Kou double-exponential jump model often implies larger distress-sensitive spreads than symmetric Gaussian-jump Merton under comparable diffusion inputs. Beyond this empirical pattern, Kou also offers theoretically desirable attributes for credit applications: asymmetric jump tails and sharper downside flexibility, which better accommodate crash-like firm-value moves when distress is relevant. However, Kou’s model failed to reflect the probability of default of a risky asset. **It is recommended to always use various models and cross-check the result for a more robust result.**

Taken together, the practical recommendation is to keep complexity aligned with conditions: use **BSM as the default baseline in mild regimes, and move to advanced models only when leverage, volatility, or market evidence indicates that tail risk is first-order.**

A central limitation remains data quality and observability. In practice we do not directly observe the true firm-asset process, true asset volatility, or full debt structural details with sufficient precision. As a result, inferred asset values and parameters are model-dependent, and estimated credit prices/spreads/PDs are inevitably biased approximations rather than ground truth. This is not unique to this report; it is a common challenge for practitioners implementing Merton-style structural models in real-world credit risk. In reality, scandal involving significant paperwork failures, misappropriation of fund, and regulatory evasion often happen without any warning from data, so the model can’t really capture the full picture of credit risk even if the relevant data are publicly observable. The recommendation can be extended to **explore model beyond Merton’s framework**, one should employ ensemble method (i.e. multiple uncorrelated models, e.g. credit score model) to detect more general warning that stochastic calculus model failed to capture.

6.1 Further Study

The following metric are common in credit risk market, often required by regulation (e.g. IFRS 9, Basel III). We skipped those metrics due to the technical challenge of obtaining the relevant data from public data source.

1. Loss Given Default (LGD). LGD is the fraction of the exposure that is not recovered if default occurs, expressed as a proportion of the amount at risk (often as a decimal in $[0, 1]$):

$$\text{LGD} = 1 - \text{recovery rate}.$$

Where the recovery rate is usually not directly observable from public data.

2. Exposure at Default (EAD). EAD is the expected exposure to the counterparty at the time of default, in monetary units:

$$\text{EAD} = \mathbb{E}[\text{exposure at default}]$$

Where the exposure is usually not directly observable from public data.

3. Expected credit loss (ECL). Under IFRS 9 Financial Instruments, the expected credit loss is calculated as:

$$\boxed{\text{ECL} = \text{PD} \times \text{LGD} \times \text{EAD}}$$

with $\text{PD}, \text{LGD} \in [0, 1]$ and EAD in currency.

6.2 Beyond Jump Models: Black–Cox and Copulas

Black–Cox. Black–Cox extends the framework by allowing default at the first hitting time of a barrier (for example, when assets fall below a covenant threshold). Economically, it adds early default and can produce more realistic short-maturity credit spreads and different term structures than a pure terminal default put. It is a natural next step when your empirical spreads at one to three years are driven by liquidity/solvency triggers, not only by the distribution of A_T . In theory, one may apply the theory of exotic knock-out option pricing (down-and-out barrier option) for such setup.

Copulas. When the goal is not a single firm but joint default risk—for example, losses on a basket, a tranche, or a portfolio—copula models separate marginal default (or loss) distributions from a dependence structure (Gaussian, t , factor copulas, etc.). Copulas are central in structured credit and portfolio risk because they let you stress correlation and tail dependence, which diffusion or even single-name jump models do not specify by themselves.

7 Appendix

7.1 Pricing algorithm for Multiple debt/coupon payment structure

The Merton model cannot price coupon-bearing debt by simply pricing each component separately. Pricing requires examining all coupon payments within a single integrated model, making assumptions about asset sales critical. (Lando, 2004)

7.1.1 Two-Coupon Example

Consider a coupon bond with two coupons D_1 and D_2 payable at dates t_1 and t_2 .

After First Coupon ($t > t_1$)

If the firm survives past t_1 with assets worth A_t , the remaining coupon D_2 can be valued using the standard Merton model:

$$B(A_t, t) = D_2 b(t, t_2) - P^{BS}(A_t, D_2, t_2 - t) \quad (85)$$

At First Coupon Date (t_1)

The situation at t_1 is more complex and depends critically on assumptions about equity owners' control over firm assets.

No Asset Sales Assumption

Equity owners cannot use firm assets to pay debt. They must finance debt payments either:

- Out of their own pockets, or
- By issuing new equity

Under no information asymmetries, both options are equivalent. If issuing M new shares to N existing shares:

$$\frac{M}{M+N} S_{t_1} = D_1 \quad (86)$$

where S_t is total equity value at time t .

This dilutes existing equity from S_{t_1} to $(N/(M+N))S_{t_1}$, causing a fall of $S_{t_1} - D_1$. Hence, paying out of pocket or dilution yield the same result.

Default condition: Equity owners pay coupon D_1 if and only if:

$$D_1 < C(A_{t_1}, D_2, t_2 - t_1) \quad (87)$$

Otherwise, default occurs and debtholders take over the firm.

7.1.2 Recursive Pricing Algorithm (No Asset Sales)

Given N coupons $D_1, \dots, D_N$ at dates $t_1, \dots, t_N$:

1. Price debt and equity at dates $t > t_{N-1}$ using standard Merton setup.
2. At t_{N-1} , find the critical value $\tilde{A}_{N-1}$ where:

$$D_{N-1} = C(\tilde{A}_{N-1}, t_N - t_{N-1}, D_N) \quad (88)$$

3. At date t_{N-1} , set boundary conditions:

$$S(A, t_{N-1}) = \begin{cases} C(A, D_N, t_N - t_{N-1}) - D_{N-1} & \text{for } A \geq \tilde{A}_{N-1} \\ 0 & \text{for } A < \tilde{A}_{N-1} \end{cases} \quad (89)$$

$$B(A, t_{N-1}) = \begin{cases} D_{N-1} + V - C(A, D_N, t_N - t_{N-1}) & \text{for } A \geq \tilde{A}_{N-1} \\ A & \text{for } A < \tilde{A}_{N-1} \end{cases} \quad (90)$$

4. Numerically value equity right after coupon payment at date t_{N-2} .
The value $\tilde{A}_{N-2}$ satisfies: equity is worth D_{N-2} right after the coupon payment at t_{N-2} .
5. Use the same procedure as in (3) to set boundary conditions at date t_{N-2} and continue recursively backwards to time 0.

7.1.3 Asset Sales Model

When asset sales are allowed, we still work recursively backwards but must adjust both the default boundary and asset value.

Optimal Strategy at t_{N-1}

If assets are worth more than D_{N-1} , default is never optimal (leaves equity with zero). Instead, equity owners should:

- Sell D_{N-1} worth of assets to cover the coupon
- Continue with assets worth $A(t_{N-1}) - D_{N-1}$
- Equity is then worth $C(A_{t_{N-1}} - D_{N-1}, D_N, t_N - t_{N-1})$

7.1.4 Comparison with Payment Out of Pocket

Paying out of pocket leaves equity with:

$$C(A_{t_{N-1}}, D_N, t_N - t_{N-1}) - D_{N-1} \quad (91)$$

But this is smaller than $C(A_{t_{N-1}} - D_{N-1}, D_N, t_N - t_{N-1})$ since the derivative of call option $\delta_A C < 1$ for all V , giving:

$$C(A_{t_{N-1}}) - C(A_{t_{N-1}} - D_{N-1}) < D_{N-1} \quad (92)$$

This is intuitively obvious: coupon payment by equity owners alone benefits both equity and debt, but is an expense to equity only.

Summary: Asset Dynamics with Sales

- Asset value follows geometric Brownian motion between coupon dates
- If asset sale finances a coupon, value drops by the coupon amount
- Equity set to zero if asset value at coupon date falls below coupon payment
- Debtholders take over the firm in that case
- If assets are sufficient, subtract coupon payment from asset value

8 Reference

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